

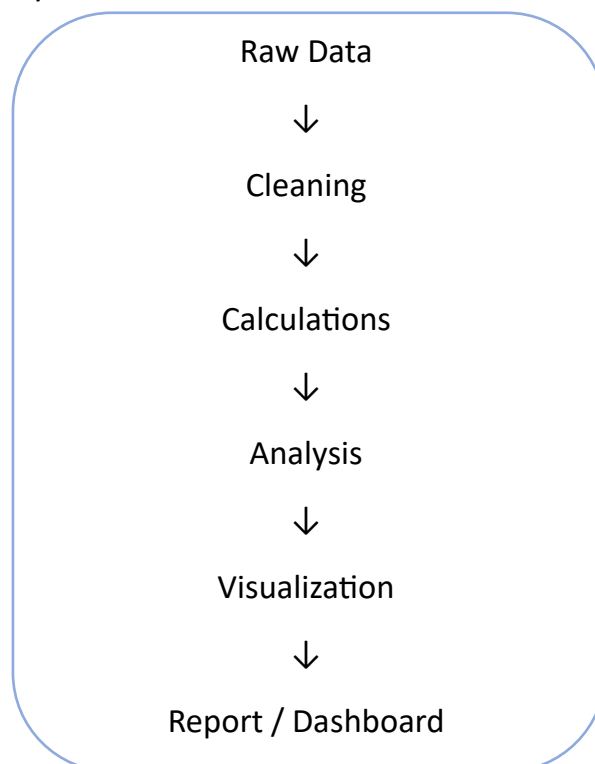
Excel Basics

1. What is Excel?

Excel is a spreadsheet application used to:

- Store data
- Clean data
- Calculate values
- Analyze data
- Create reports
- Create charts
- Build dashboards
- Automate repetitive work

For a **Data Analyst**, Excel is mainly used for:



2. Excel Terminology

You must know these terms.

*Workbook

An Excel file.

*Worksheet

A sheet inside a workbook.

*Cell

Intersection of a row and column.

*Row

Horizontal.

*Column

Vertical.

*Cell Range

Multiple cells.

A1:A10

means A1 through A10.

3. Data Types

As a Data Analyst, understand what type of data you're working with.

*Text

Pranali | Mumbai | Laptop | IT

*Numbers

100 | 25000 | 89.5

*Dates

15-08-2026 | 01-Jan-2025

*Time

10:30 AM | 18:45

*Boolean

TRUE | FALSE

*Formula

=A2+B2

4. Create Your First Dataset

Don't just read the notes. **Open Excel and type this dataset yourself.**

Employee Dataset

Employee_ID	Name	Department	City	Salary	Experience
E001	Rahul	IT	Pune	45000	2
E002	Priya	HR	Mumbai	38000	3
E003	Amit	IT	Nashik	52000	4
E004	Sneha	Finance	Pune	48000	3
E005	Rohan	Marketing	Mumbai	42000	2
E006	Neha	IT	Mumbai	60000	5
E007	Akash	Finance	Nashik	55000	4
E008	Pooja	HR	Pune	35000	1
E009	Karan	Marketing	Nashik	46000	3
E010	Anjali	IT	Pune	58000	4

Enter it starting from **A1**.

5. Formatting

Select your dataset and practice:

***Header formatting**

Make:

Employee_ID	Name	Department	City	Salary	Experience
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Practice:

- Bold
- Font size
- Alignment
- Borders
- Number formatting
- Column width
- Row height

*Salary formatting

Salary should look like:

45,000 | 38,000 | 52,000

instead of:

45000 | 38000 | 52000

6. Column Width

If you see:

#####

don't panic 😊

It usually means the column isn't wide enough to display the value.

Double-click the boundary between column letters.

Example:

A | B | C

Double-click between **C and D**.

Excel will AutoFit the column.

7. Basic Cell References

This is extremely important because formulas will use cell references.

Suppose:

A1 = 100

B1 = 200

Then:

=A1+B1

returns:

300

Common references

A1

B5

C10

D25

8. Basic Arithmetic

Excel can work like a calculator.

*Addition

=10+20

Result: 30

*Subtraction

=50-20

Result: 30

*Multiplication

=10*5

Result: 50

*Division

=100/5

Result: 20

*Power

=5^2

Result: 25

9. Formula Basics

Every Excel formula starts with =.

Example:

=10+20

or:

=A2+B2

or:

=C2*D2

Example

Suppose:

Product	Quantity	Price
Laptop	2	50000

In D2:

=B2*C2

Result: 100000

10. Relative Cell Reference

Suppose:

A	B	C
Product	Quantity	Price
Laptop	2	50000
Mouse	5	500
Keyboard	3	1500

In D2:

=B2*C2

Then drag the formula down.

Excel automatically changes:

=B2*C2

=B3*C3

=B4*C4

This is called a **relative reference**.

You'll use this constantly in data analysis.

11. AutoFill

Instead of writing the same formula repeatedly:

1. Enter formula in first cell.
2. Select the cell.
3. Move cursor to bottom-right corner.
4. Drag down.

Example:

=B2*C2

becomes:

=B3*C3

=B4*C4

=B5*C5

12. Sort Data

Sorting means arranging data in an order.

Excel supports sorting text, numbers, dates and multiple columns.

Using our Employee Dataset:

Sort Salary Low → High

Data → Sort → Salary → Smallest to Largest

Sort Name A → Z

Data Analyst use case

"Show me the employees with the highest salaries."

Sort Salary from largest to smallest.

13. Filter Data

Filtering means showing only the rows that satisfy your condition. Excel can filter on one or multiple columns.

Example:

Filter:

Department = IT

You'll see:

Employee	Department
Rahul	IT
Amit	IT
Neha	IT
Anjali	IT

Other rows are temporarily hidden.

Another example

Filter:

City = Pune

You'll see only Pune employees.

14. Multiple Filters

This is important.

Suppose:

Department = IT

City = Pune

Excel will show employees satisfying **both conditions**.

From our dataset:

Rahul

Anjali

This is the beginning of analytical thinking:

"Find IT employees working in Pune."

15. Freeze Panes

When a dataset has thousands of rows, you don't want your headers to disappear while scrolling.

Use:

View → Freeze Panes → Freeze Top Row

Example:

Employee_ID | Name | Department | Salary

E001

E002

E003

...

E1000

When you scroll down, the headers remain visible.

16. Excel Tables

This is your first **proper Data Analyst feature**.

Select your dataset and press:

Ctrl + T

Or:

Home → Format as Table

Excel tables make data easier to filter and analyze.

Select:

My table has headers

Now you'll get filter buttons in the headers.

Example:

Employee_ID ▼

Name ▼

Department ▼

City ▼

Salary ▼

Experience ▼

17. Why Tables Matter for Data Analysts

Normal range:

A1:F11

Table:

EmployeeTable

Tables automatically support features such as filtering and structured data organization, and they work well as sources for later analysis.

You'll use tables later with:

- PivotTables
- Power Query
- Charts
- Formulas
- Dashboards

18. Important Keyboard Shortcuts

Start memorizing these.

Shortcut	Use
Ctrl + C	Copy
Ctrl + V	Paste
Ctrl + X	Cut
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + S	Save
Ctrl + F	Find
Ctrl + H	Replace
Ctrl + A	Select
Ctrl + T	Create Table
Ctrl + 1	Format Cells
Ctrl + Arrow	Jump through data
Ctrl + Shift + Arrow	Select data range
F2	Edit cell
Alt + =	AutoSum